

1991 年数学(三) 真题解析

一、填空题

(1) 【答案】 $\cos(xy) \cdot e^{\sin(xy)}(ydx + xdy)$.

【解】 由 $\frac{\partial z}{\partial x} = y \cos(xy) \cdot e^{\sin(xy)}$, $\frac{\partial z}{\partial y} = x \cos(xy) \cdot e^{\sin(xy)}$ 得

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy = \cos(xy) \cdot e^{\sin(xy)}(ydx + xdy).$$

(2) 【答案】 $-1, -1, 1$.

【解】 $f'(x) = 3x^2 + a, g'(x) = 2bx$,

$$\text{由题意得} \begin{cases} -1 - a = 0, \\ b + c = 0, \\ 3 + a = -2b, \end{cases}$$

解得 $a = -1, b = -1, c = 1$.

(3) 【答案】 $-(n+1), -\frac{1}{e^{n+1}}$.

【解】 $f'(x) = (x+1)e^x, f''(x) = (x+2)e^x$, 由归纳法得 $f^{(n)}(x) = (x+n)e^x$,

由 $f^{(n+1)}(x) = (x+n+1)e^x = 0$ 得 $x = -(n+1)$,

当 $x < -(n+1)$ 时, $f^{(n+1)}(x) < 0$; 当 $x > -(n+1)$ 时, $f^{(n+1)}(x) > 0$,

故当 $x = -(n+1)$ 时, $f^{(n)}(x)$ 取极小值, 且极小值为 $f^{(n)}(-n-1) = -\frac{1}{e^{n+1}}$.

(4) 【答案】 $\begin{pmatrix} O & B^{-1} \\ A^{-1} & O \end{pmatrix}$.

【解】 令 $X^{-1} = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}$,

$$\text{由} \begin{pmatrix} O & A \\ B & O \end{pmatrix} \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix} = \begin{pmatrix} E & O \\ O & E \end{pmatrix} \text{得} \begin{cases} AX_{21} = E, \\ AX_{22} = O, \\ BX_{11} = O, \\ BX_{12} = E. \end{cases}$$

从而 $X_{11} = O, X_{12} = B^{-1}, X_{21} = A^{-1}, X_{22} = O$, 故 $X^{-1} = \begin{pmatrix} O & B^{-1} \\ A^{-1} & O \end{pmatrix}$.

(5) 【答案】 $X \sim \begin{pmatrix} -1 & 1 & 3 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$.

【解】 因为 $F(x)$ 的图形为阶梯形, 所以随机变量 X 为离散型, 其可能的取值为 $-1, 1, 3$,

$$P\{X = -1\} = F(-1) - F(-1-0) = 0.4,$$

$$P\{X = 1\} = F(1) - F(1-0) = 0.8 - 0.4 = 0.4,$$

$$P\{X = 3\} = F(3) - F(3-0) = 1 - 0.8 = 0.2,$$

故 X 的概率分布为 $X \sim \begin{pmatrix} -1 & 1 & 3 \\ 0.4 & 0.4 & 0.2 \end{pmatrix}$.

二、选择题

(1) 【答案】 (A).

$$\text{【解】 由 } \lim_{x \rightarrow 0^+} \left(1 + \frac{1}{x}\right)^x = e^{\lim_{x \rightarrow 0^+} x \ln \left(1 + \frac{1}{x}\right)} = e^{\lim_{x \rightarrow 0^+} \frac{\ln \left(1 + \frac{1}{x}\right)}{\frac{1}{x}}} = e^{\lim_{t \rightarrow +\infty} \frac{\ln(1+t)}{t}} = e^{\lim_{t \rightarrow +\infty} \frac{1}{t+1}} = e^0 = 1,$$

故选(A).

(2) 【答案】 (D).

$$\text{【解】 由 } 0 \leq a_n < \frac{1}{n} \text{ 得 } 0 \leq |(-1)^n a_n^2| = a_n^2 < \frac{1}{n^2},$$

因为 $\sum_{n=1}^{\infty} \frac{1}{n^2}$ 收敛, 所以 $\sum_{n=1}^{\infty} |(-1)^n a_n^2|$ 收敛, 即级数 $\sum_{n=1}^{\infty} (-1)^n a_n^2$ 绝对收敛, 故 $\sum_{n=1}^{\infty} (-1)^n a_n^2$ 一定收敛, 故选(D).

(3) 【答案】 (B).

$$\text{【解】 由 } \mathbf{A}\mathbf{X} = \lambda\mathbf{X} \text{ 得 } \mathbf{A}^{-1}\mathbf{X} = \lambda^{-1}\mathbf{X}.$$

$$\text{而 } \mathbf{A}^* = |\mathbf{A}| \mathbf{A}^{-1}, \text{ 于是 } \mathbf{A}^* \mathbf{X} = |\mathbf{A}| \mathbf{A}^{-1} \mathbf{X} = \lambda^{-1} |\mathbf{A}| \mathbf{X},$$

故 $\mathbf{A}^*$ 的一个特征根为 $\lambda^{-1} |\mathbf{A}|$, 故选(B).

(4) 【答案】 (D).

$$\text{【解】 因为 } A \text{ 和 } B \text{ 不相容, 所以 } AB = \emptyset,$$

$$\text{于是 } P(A-B) = P(A) - P(AB) = P(A), \text{ 故选(D).}$$

(5) 【答案】 (B).

$$\text{【解】 由 } E(XY) = E(X) \cdot E(Y) \text{ 得 } \text{Cov}(X, Y) = 0,$$

$$\text{于是 } D(X+Y) = D(X) + D(Y) + 2\text{Cov}(X, Y) = D(X) + D(Y), \text{ 故选(B).}$$

三、【解】

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{e^x + e^{2x} + \cdots + e^{nx}}{n} \right)^{\frac{1}{x}} &= \lim_{x \rightarrow 0} \left[\left(1 + \frac{e^x + e^{2x} + \cdots + e^{nx} - n}{n} \right)^{\frac{1}{\frac{e^x + e^{2x} + \cdots + e^{nx} - n}{n}}} \right]^{\frac{1}{x} \cdot \frac{e^x + e^{2x} + \cdots + e^{nx} - n}{n}} \\ &= e^{\lim_{x \rightarrow 0} \frac{1}{x} \cdot \frac{e^x + e^{2x} + \cdots + e^{nx} - n}{n}} = e^{\frac{1}{n} \lim_{x \rightarrow 0} \frac{e^x + e^{2x} + \cdots + e^{nx} - n}{x}} \\ &= e^{\frac{1}{n} \lim_{x \rightarrow 0} (e^x + 2e^{2x} + \cdots + ne^{nx})} = e^{\frac{1+2+\cdots+n}{n}} = e^{\frac{n+1}{2}}. \end{aligned}$$

$$\text{四、【解】 由 } \sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1 \text{ 得 } y = b \left(1 - \sqrt{\frac{x}{a}} \right)^2 = b \left(1 - \frac{2}{\sqrt{a}} \sqrt{x} + \frac{x}{a} \right),$$

$$\text{于是 } \iint_D y \, dx \, dy = \int_0^a dx \int_0^{b \left(1 - \frac{2}{\sqrt{a}} \sqrt{x} + \frac{x}{a} \right)} y \, dy$$

$$\begin{aligned} &= \frac{b^2}{2} \int_0^a \left(\sqrt{\frac{x}{a}} - 1 \right)^4 dx \stackrel{\sqrt{\frac{x}{a}} = t}{=} \frac{b^2}{2} \int_0^1 (t-1)^4 \cdot 2at \, dt \\ &= ab^2 \int_0^1 (t-1)^4 \cdot [(t-1) + 1] \, dt \\ &= ab^2 \left[\frac{1}{6} (t-1)^6 \Big|_0^1 + \frac{1}{5} (t-1)^5 \Big|_0^1 \right] \\ &= ab^2 \left(\frac{1}{5} - \frac{1}{6} \right) = \frac{ab^2}{30}. \end{aligned}$$

五、【解】 方法一 由 $xy \frac{dy}{dx} = x^2 + y^2$ 得 $\frac{dy}{dx} = \frac{x}{y} + \frac{y}{x}$,

令 $u = \frac{y}{x}$, 则

$$u + x \frac{du}{dx} = \frac{1}{u} + u,$$

整理得 $u du = \frac{dx}{x}$, 积分得

$$u^2 = \ln x^2 + C,$$

将 $y|_{x=e} = 2e$ 代入得 $C = 2$, 所求的特解为 $y^2 = x^2 \ln x^2 + 2x^2$.

方法二 由 $xy \frac{dy}{dx} = x^2 + y^2$ 得 $\frac{x}{2} \frac{d(y^2)}{dx} = x^2 + y^2$,

令 $y^2 = u$, 则 $\frac{du}{dx} - \frac{2}{x}u = 2x$,

解得

$$u = \left(\int 2x \cdot e^{\int -\frac{2}{x} dx} dx + C \right) \cdot e^{-\int -\frac{2}{x} dx} = (2 \ln x + C)x^2,$$

由 $y|_{x=e} = 2e$ 得 $C = 2$, 即 $y^2 = 2(\ln x + 1)x^2$.

六、【解】 如右图所示,

$$\text{由 } \begin{cases} y = 1 - x^2, \\ y = ax^2 \end{cases} \text{ 得 } x = \frac{1}{\sqrt{a+1}}.$$

曲线 $L_1: y = 1 - x^2$ ($0 \leq x \leq 1$) 与两坐标轴围成的面积为

$$S = \int_0^1 (1 - x^2) dx = \frac{2}{3},$$

曲线 $L_1: y = 1 - x^2$ 与 $L_2: y = ax^2$ 围成的位于第一象限上方的面积为

$$\begin{aligned} S_1 &= \int_0^{\frac{1}{\sqrt{a+1}}} (1 - x^2 - ax^2) dx \\ &= \frac{1}{\sqrt{a+1}} - \frac{(a+1)}{3} x^3 \Big|_0^{\frac{1}{\sqrt{a+1}}} = \frac{2}{3\sqrt{a+1}}, \end{aligned}$$

由 $2S_1 = S$ 得 $\frac{4}{3\sqrt{a+1}} = \frac{2}{3}$ 得 $a = 3$.

七、【解】 收入函数为

$$R = p_1 q_1 + p_2 q_2 = 24p_1 - 0.2p_1^2 + 10p_2 - 0.05p_2^2,$$

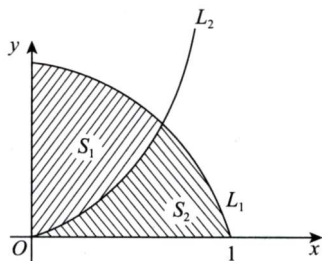
利润函数为

$$\begin{aligned} L &= R - C = p_1 q_1 + p_2 q_2 = 24p_1 - 0.2p_1^2 + 10p_2 - 0.05p_2^2 - [35 + 40(q_1 + q_2)] \\ &= 32p_1 - 0.2p_1^2 + 12p_2 - 0.05p_2^2 - 1395, \end{aligned}$$

$$\text{由 } \begin{cases} \frac{\partial L}{\partial p_1} = 32 - 0.4p_1 = 0, \\ \frac{\partial L}{\partial p_2} = 12 - 0.1p_2 = 0 \end{cases} \text{ 得 } p_1 = 80, p_2 = 120,$$

故当 $p_1 = 80, p_2 = 120$ 时, 总利润最大, 最大利润为

$$L_{\max} = 605.$$



六题图

八、【证明】 $f(x) = e^{x \ln(1+\frac{1}{x})}$,

$$f'(x) = e^{x \ln(1+\frac{1}{x})} \cdot \left[\ln\left(1+\frac{1}{x}\right) + x \cdot \frac{1}{1+\frac{1}{x}} \cdot \left(-\frac{1}{x^2}\right) \right]$$

$$= \left(1+\frac{1}{x}\right)^x \cdot \left[\ln\left(1+\frac{1}{x}\right) - \frac{1}{x+1} \right],$$

$$\text{记 } g(x) = \ln\left(1+\frac{1}{x}\right) - \frac{1}{x+1}, \quad g'(x) = \frac{-1}{x(1+x)^2} < 0 (x > 0),$$

所以 $g(x)$ 在 $(0, +\infty)$ 上单调减少.

因为 $\lim_{x \rightarrow +\infty} g(x) = 0$, 所以对任意 $x \in (0, +\infty)$, 有 $g(x) > 0$,

于是当 $x > 0$ 时, $f'(x) > 0$, 即函数 $f(x) = \left(1+\frac{1}{x}\right)^x$ 在 $(0, +\infty)$ 内单调增加.

九、【解】 令 $x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3 = \beta$,

$$|A| = |\alpha_1, \alpha_2, \alpha_3| = (\lambda + 3)\lambda^2,$$

(1) 当 $\lambda \neq -3, \lambda \neq 0$ 时, 方程组 $x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3 = \beta$ 有唯一解,

即 β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 且表达式唯一;

(2) 当 $\lambda = 0$ 时, $r(A) = r(\bar{A}) = 1 < 3$,

方程组 $x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3 = \beta$ 有无数个解,

即 β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 且表达式不唯一;

(3) 当 $\lambda = -3$ 时,

$$\bar{A} = \begin{pmatrix} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & -3 \\ 1 & 1 & -2 & 9 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 1 & -3 \\ 0 & -3 & 3 & -6 \\ 0 & 3 & -3 & 12 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -2 & 1 & -3 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 6 \end{pmatrix},$$

由 $r(A) \neq r(\bar{A})$ 得方程组 $x_1 \alpha_1 + x_2 \alpha_2 + x_3 \alpha_3 = \beta$ 无解,

即 β 不可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示.

十、【解】 令 $A = \begin{pmatrix} 1 & \lambda & -1 \\ \lambda & 4 & 2 \\ -1 & 2 & 4 \end{pmatrix}, X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$, 则 $f = X^T A X$,

二次型 $f = X^T A X$ 为正定二次型的充分必要条件是

$$1 > 0, \quad \begin{vmatrix} 1 & \lambda \\ \lambda & 4 \end{vmatrix} > 0, \quad |A| = \begin{vmatrix} 1 & \lambda & -1 \\ \lambda & 4 & 2 \\ -1 & 2 & 4 \end{vmatrix} > 0,$$

解得 $-2 < \lambda < 1$.

十一、【证明】 令 $A = (\alpha_1, \alpha_2, \dots, \alpha_n)$, $A^T A =$

$$\begin{pmatrix} \alpha_1^T \alpha_1 & \alpha_1^T \alpha_2 & \cdots & \alpha_1^T \alpha_n \\ \alpha_2^T \alpha_1 & \alpha_2^T \alpha_2 & \cdots & \alpha_2^T \alpha_n \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_n^T \alpha_1 & \alpha_n^T \alpha_2 & \cdots & \alpha_n^T \alpha_n \end{pmatrix},$$

则向量组 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关的充分必要条件是 $r(A) = n$,

因为 $r(A) = r(A^T A)$, 所以 $\alpha_1, \alpha_2, \dots, \alpha_n$ 线性无关的充分必要条件是 $r(A^T A) = n$,

即 $|A^T A| \neq 0$.

十二、【解】 X 的可能取值为 $0, 1, 2, 3$,

$$P\{X=0\} = \frac{1}{2},$$

$$P\{X=1\} = \frac{1}{2} \times \left(1 - \frac{1}{2}\right) = \frac{1}{4},$$

$$P\{X=2\} = \frac{1}{2} \times \frac{1}{2} \times \left(1 - \frac{1}{2}\right) = \frac{1}{8},$$

$$P\{X=3\} = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8},$$

$$\text{则 } X \sim \begin{pmatrix} 0 & 1 & 2 & 3 \\ \frac{1}{2} & \frac{1}{4} & \frac{1}{8} & \frac{1}{8} \end{pmatrix}.$$

十三、【解】 (X, Y) 的联合密度函数为

$$f(x, y) = \begin{cases} \frac{1}{\pi r^2}, & x^2 + y^2 \leq r^2, \\ 0, & x^2 + y^2 > r^2, \end{cases}$$

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy,$$

$$\text{当 } |x| \geq r \text{ 时, } f_X(x) = 0;$$

$$\text{当 } |x| < r \text{ 时, } f_X(x) = \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \frac{1}{\pi r^2} dy = \frac{2\sqrt{r^2-x^2}}{\pi r^2};$$

$$\text{即 } f_X(x) = \begin{cases} \frac{2\sqrt{r^2-x^2}}{\pi r^2}, & |x| < r, \\ 0, & |x| \geq r, \end{cases}$$

$$\text{同理 } f_Y(y) = \begin{cases} \frac{2\sqrt{r^2-y^2}}{\pi r^2}, & |y| < r, \\ 0, & |y| \geq r, \end{cases}$$

$$(1) E(X) = \int_{-r}^r x f_X(x) dx = 0, \quad E(Y) = \int_{-r}^r y f_Y(y) dy = 0,$$

$$E(XY) = \iint_{x^2+y^2 \leq r^2} xy f(x, y) dx dy = 0,$$

$$\text{由 } \text{Cov}(X, Y) = E(XY) - E(X)E(Y) = 0 \text{ 得 } X, Y \text{ 的相关系数 } \rho = 0.$$

$$(2) \text{ 因为 } f(x, y) \neq f_X(x)f_Y(y), \text{ 所以 } X, Y \text{ 不独立.}$$

十四、【解】 $L(\lambda) = f(x_1; \lambda)f(x_2; \lambda) \cdots f(x_n; \lambda)$

$$= \lambda^n a^n (x_1 x_2 \cdots x_n)^{a-1} e^{-\lambda(x_1^a + x_2^a + \cdots + x_n^a)} \quad (x_i > 0, i = 1, 2, \cdots, n),$$

$$\ln L(\lambda) = n \ln \lambda + n \ln a + (a-1) \ln(x_1 x_2 \cdots x_n) - \lambda(x_1^a + x_2^a + \cdots + x_n^a),$$

$$\text{由 } \frac{d}{d\lambda} \ln L(\lambda) = \frac{n}{\lambda} - (x_1^a + x_2^a + \cdots + x_n^a) = 0 \text{ 得 } \lambda \text{ 的最大似然估计值为 } \lambda = \frac{n}{x_1^a + x_2^a + \cdots + x_n^a},$$

$$\text{故参数 } \lambda \text{ 的最大似然估计量为 } \hat{\lambda} = \frac{n}{X_1^a + X_2^a + \cdots + X_n^a}.$$